



Which Method Should I Use?

A practical decision guide for orientation data in archaeoastronomy and ethnoastronomy

SEAC 2026 Workshop: Circular Statistics in Archaeoastronomy and Ethnoastronomy Using R



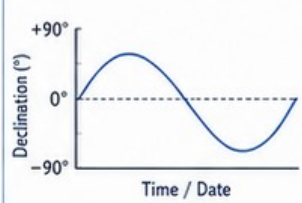
1 Step 1 - What kind of data do I have?



Directions or azimuths
Use circular statistics.



Axes with no meaningful start/end direction
Treat them as axial data before applying directional methods.

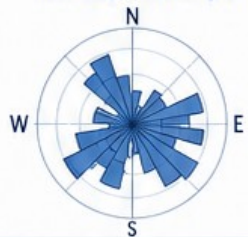


Astronomical declinations
For the workshop curvigram, declination is treated as a linear angular coordinate over the analysed range.

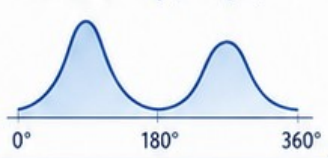
2 Step 2 - What question am I asking?

	Research question	First workshop method	What it tells you	Main caution
1	What does the pattern look like?	Individual-direction plot / rose diagram	Shape, modes, spread	Appearance depends partly on graphical choices
2	What is the typical direction?	Circular mean	Central direction	Can mislead for multimodal data
3	How concentrated are directions?	R-bar	Net directional concentration	Low R-bar can result from opposing clusters
4	How dispersed are directions?	Circular variance / circular SD	Circular spread	Interpret together with the plot
5	How precise is the mean direction?	Bootstrap CI	Precision of estimated mean	Not archaeological certainty
6	Is there a unimodal preferred direction rather than uniformity?	Rayleigh test	Evidence against uniformity for a unimodal alternative	Can miss bimodal/multimodal structure
7	Is one bell-like circular model useful?	von Mises model	Mean direction μ and concentration κ	Do not force on obvious multimodality
8	How do uncertain declinations combine into a continuous curve?	Curvigram	Smoothed aggregate pattern	Kernel and bandwidth affect details
9	Is a curvigram feature robust?	Sensitivity analysis	Stability across reasonable settings	Not a formal significance test
10	Do two subgroups look different?	Separate plots and summaries	Descriptive differences	Formal between-group inference is outside this workshop

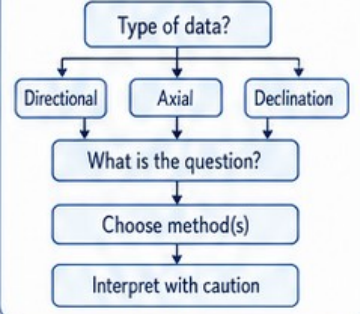
Rose diagram (example)



Circular density (example)



Decision tree motif



3 Step 3 - Recommended workflow

1 Check the data

- units;
- geographic convention;
- directional versus axial;
- missing values;
- sample size;
- measurement uncertainty.

2 Plot before testing

- unimodal?
- bimodal?
- multimodal?
- near-uniform?
- isolated observations?

3 Describe

- n ;
- mean direction if meaningful;
- R-bar;
- circular variance / SD;
- confidence interval where appropriate.

4 Test the question you actually have

- Rayleigh for a simple unimodal preferred-direction question;
- do not treat it as a universal random/non-random button.

5 Model only when useful

- a single von Mises distribution is appropriate only when one concentrated mode is scientifically plausible.

6 Represent uncertainty explicitly

- in the curvigram exercise, each observation has its own reported plus/minus uncertainty.

7 Test robustness

- change kernel and bandwidth deliberately;
- report the specifications;
- trust stable features more than specification-sensitive details.

4 Step 4 - Minimal R cookbook

```
library(circular)
az <- circular(
  dat$azimuth_deg,
  units = "degrees",
  template = "geographics",
  modulo = "2pi"
)
rose.diag(az, bins = 18)
as.numeric(mean(az)) %% 360
rho.circular(az)
1 - rho.circular(az)
rayleigh.test(az)
mle.vonmises(az)
```



5 Step 5 - Three rules

1 Plot first, test second.

2 Match the method to the question and distribution shape.

3 Statistical regularity does not automatically establish cultural intention or an astronomical target.